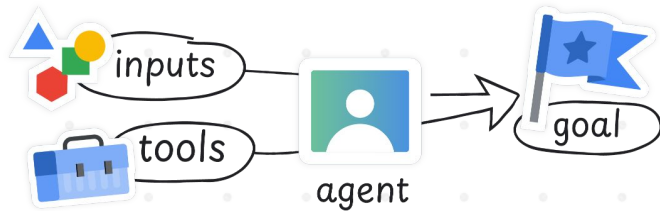


Customer Experience Agent Studio: Fundamentals Review

This course summary is your review sheet. Print or save it as a reference as you continue your learning journey.

An **agent** is a piece of software that learns how to best achieve a goal based on inputs and tools available to it.



Core components of an agent



Model: The underlying large language model (LLM) that powers understanding and response generation.

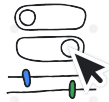


Tools: Extensions that connect the agent to external systems (APIs and data stores) to perform actions.



Reasoning loop: The iterative process where the agent considers its goal, available tools, and instructions to make decisions.

The best modern agents blend two types of logic to balance flexibility with control:

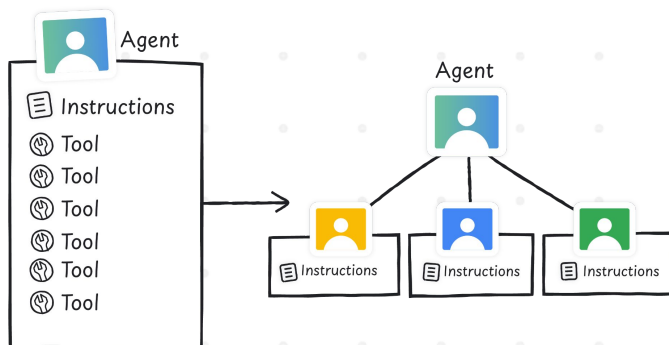


Deterministic (rule-based): Follows strict paths and logic. Best for compliance, transactions, and high-stakes logic.



Generative (AI-driven): Handles open-ended conversations and ambiguity. Best for open-ended conversations and personalized recommendations..

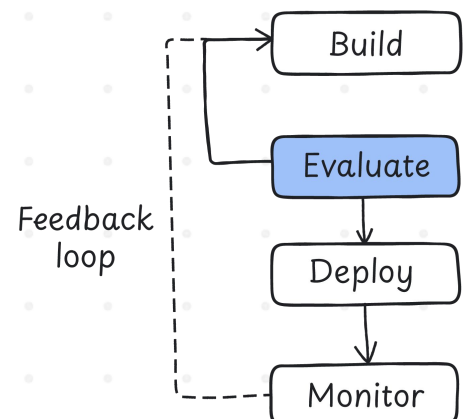
Multi-agent architectures organize complex applications like a corporate organizational chart to improve clarity and maintenance.



- **Root agent:** The "manager" that routes traffic based on user intent.
- **Sub-agents:** The "specialists" dedicated to specific domains (e.g., Billing, Tech Support).
- **When to split:** Use sub-agents when tasks are complex, conversational, or stateful. Use single agents for simple, atomic tasks.

Building agents is an iterative process.

Agent Building journey



Customer Experience Agent Studio: Fundamentals Review

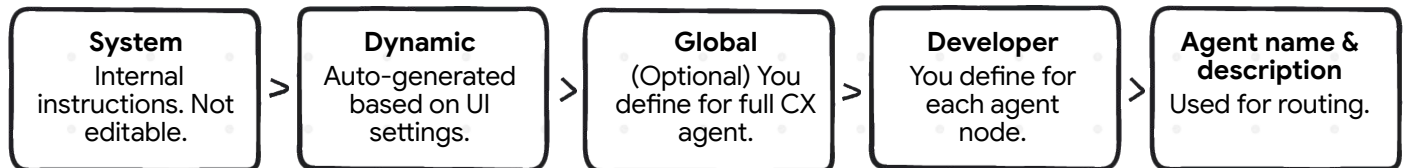
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CX Agent Studio: A managed, low-code platform that combines the conversational fluidity of generative AI with the precision of deterministic business logic to build enterprise-ready agents.

BUILD

Instructions: Detailed guidance for the agent. Appended top-down to the query:



(Tip: Type @ to link tools/agents, { to link variables)

Tools: Enable agents to take action.

- **System tools:** Built-in tools for creating a custom response or ending a session.
- **Custom tools:** Python, OpenAPI, Client, Data Store, Salesforce, ServiceNow, etc.

Variables: Store and reuse data across turns.

Global settings

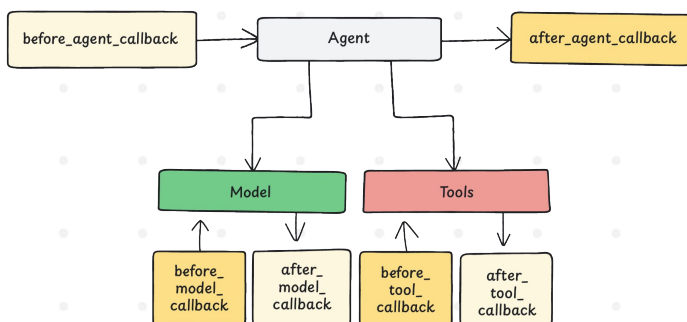
- **Voice:** Control accent, speed, and pitch.
- **Language:** Set primary and fallback language.
- **Ambient noise:** Use background audio.
- **Behavior:** Control interruptibility and patience.

Guardrails and sensitive data protection:

- **Prompt guard:** Anti-jailbreaking.
- **Blocklist:** Block words and phrases.
- **Safety:** Enforce responsible AI practices.
- **Rules:** Custom logic for specific needs.
- **DLP:** Redacts sensitive data (PII) using inspect/deidentify templates.

Callbacks: Deterministic Python code that runs at specific triggers. There are six types:

- **Agent (Pre/Post):** Setup resources / Cleanup and update final response.
- **Model (Pre/Post):** Modify prompt / Format response and error handling.
- **Tool (Pre/Post):** Validate inputs / Process outputs and save to variables.



EVALUATE AND MONITOR

Version: Snapshot in time of the agent application at that time.

Changelog: An audit trail of who did what and what changes were made.

Preview Agent: Simulator for text/voice/image testing. Also view **Conversation history**.

The trace: Step-by-step view of latency, tool calls, and variable states.

Logging

- **Conversation logging:** Conversational Insights for production traffic.
- **Audio file logging:** Raw audio in Cloud Storage.
- **Bigquery export:** Structured data for analytics.
- **Cloud logging:** Design and runtime audits.

Evaluations

- **Golden:** Strict "answer keys" for regression testing specific inputs/outputs.
- **Scenario:** AI-generated variations to test ability to handle complex goals.

DEPLOY

- **Web:** Widget for browser/mobile.
- **Telephony:** Twilio, Google Telephony Platform, and CCaaS.
- **API:** Headless access for custom apps.